NETFLIX DATA ANALYSIS USING PYTHON

Auspify Technologies Internship Project | Tasks 1–6

Executive Summary

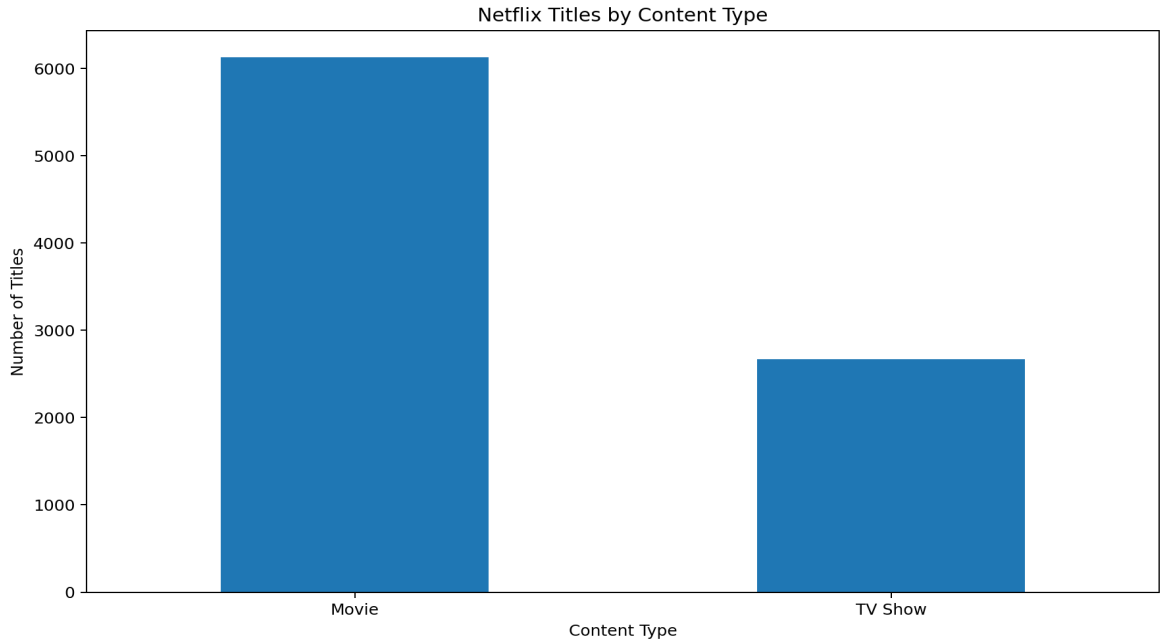
The supplied dataset contains 8,790 rows and 10 columns before cleaning. The prepared dataset contains **8,790 records**: **6,126 Movies (69.7%)** and **2,664 TV Shows (30.3%)**.

Task 1 — Data Cleaning & Preparation

Column names and text fields were standardized; type labels were normalized while preserving the official 'TV Show' spelling; rating values were uppercased; date_added was parsed as a date and release_year as a nullable integer. Exact duplicate rows and duplicate show IDs were checked. Missing descriptive categorical values are labelled 'Not Specified' instead of silently dropping records.

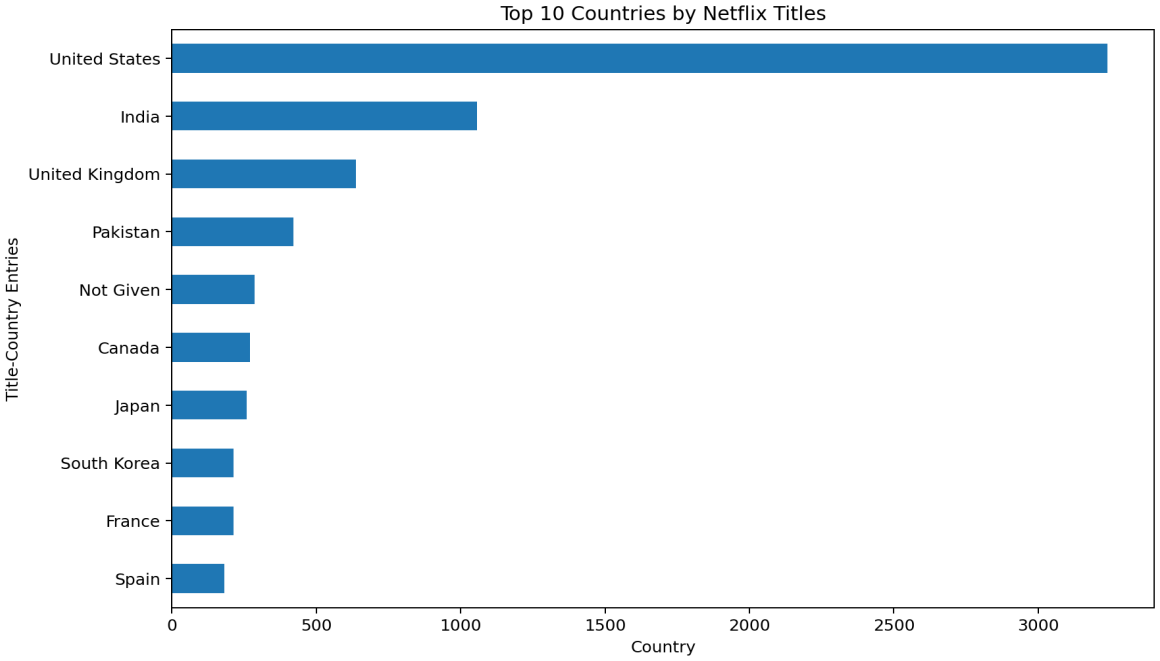
Task 2 — Content Type Analysis

Movies account for 69.7% of the catalogue (6,126 titles), while TV Shows account for 30.3% (2,664 titles).



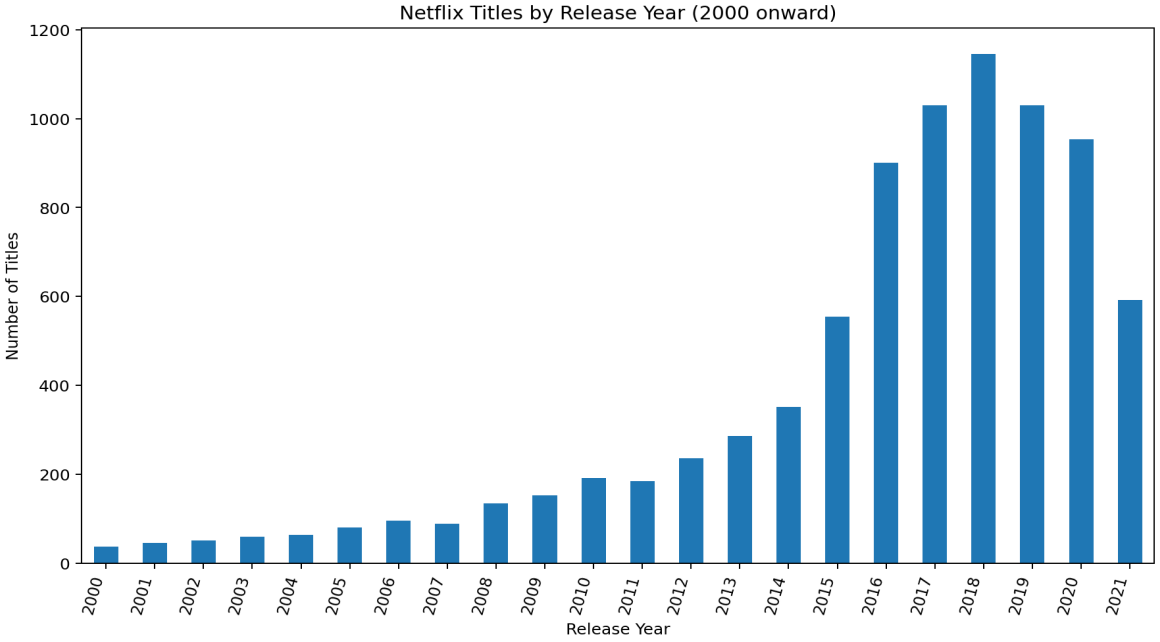
Task 3 — Country-Wise Content Analysis

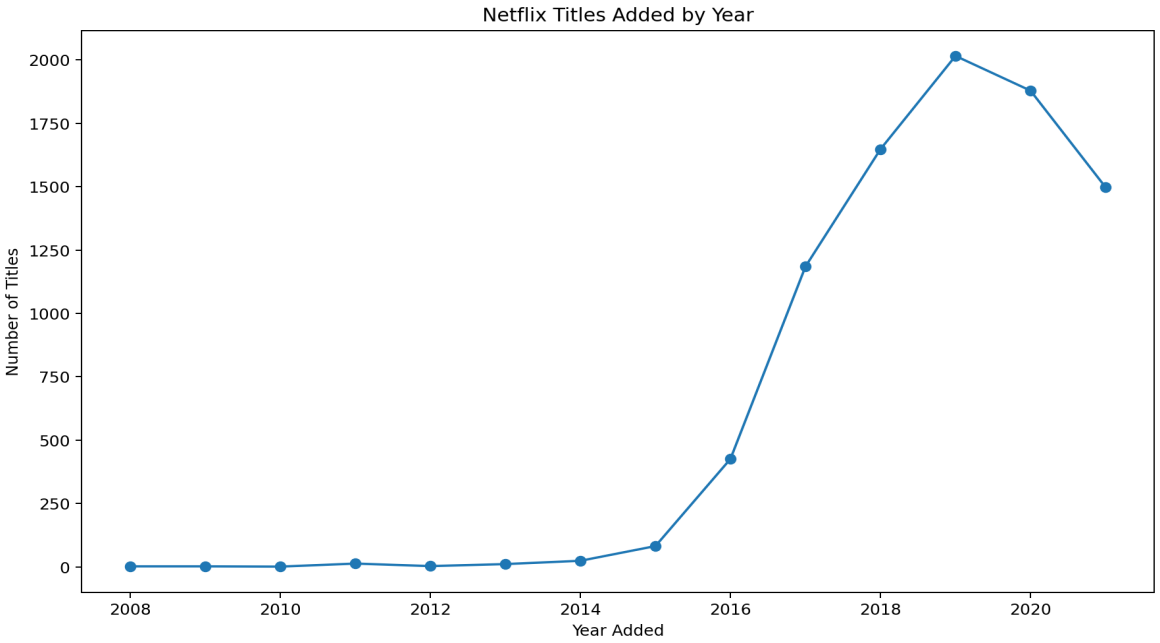
The highest country entry count is United States (3,240 title-country entries). A title listing multiple countries is counted in each listed country.



Task 4 — Trend Analysis by Release Year

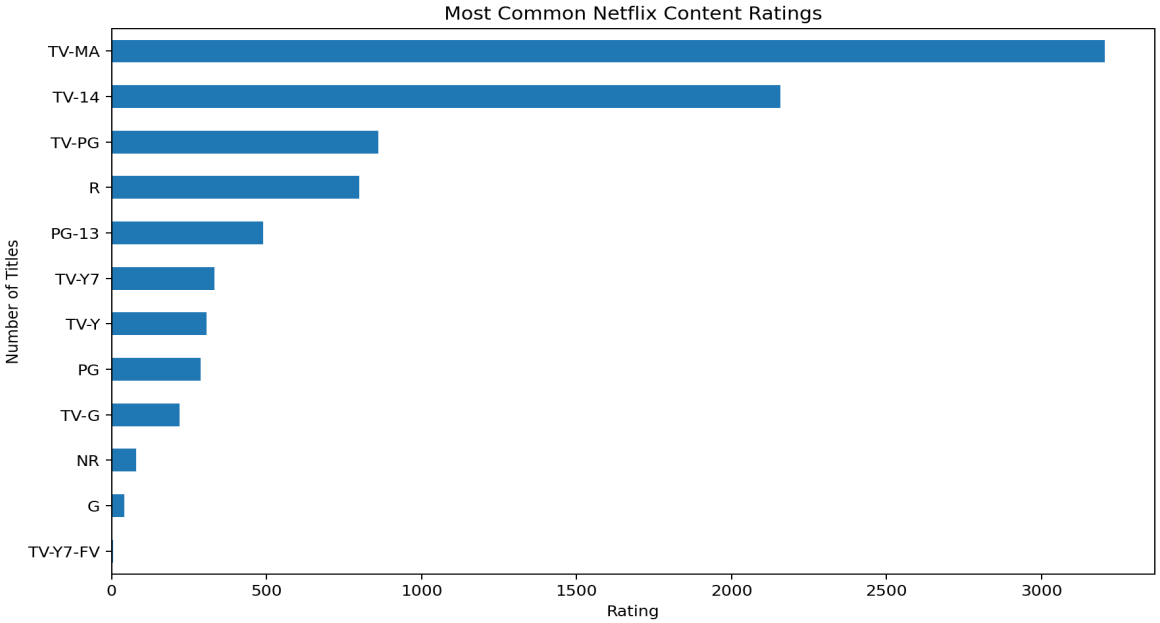
The release-year series peaks at 2018 with 1,146 titles in this dataset. The release-year chart focuses on 2000 onward for readability. The separate date-added chart describes when catalogue entries were added, not when the content was released.

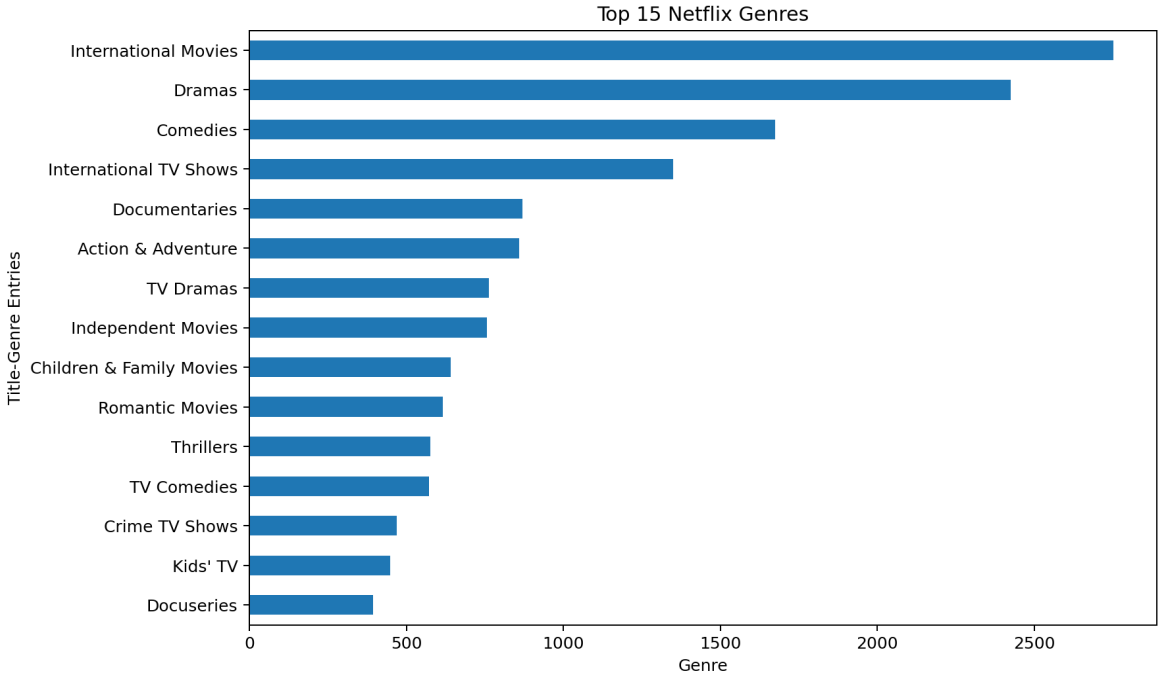


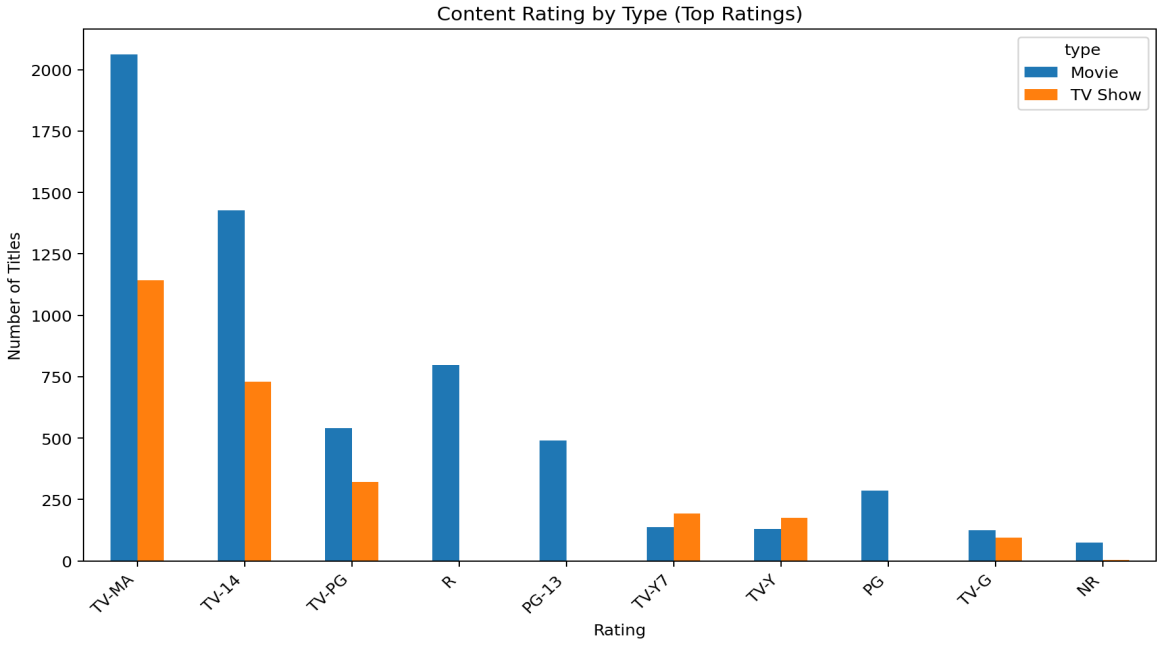


Task 5 — Rating & Genre Analysis

The most common rating label is TV-MA (3,205 titles). The most frequent genre entry is International Movies (2,752 title-genre entries). Genre and country fields are multi-label, so entries can count under more than one category.







Task 6 — Business Insights & Recommendations

• Content mix: Movies form 69.7% of this supplied catalogue, compared with 30.3% TV Shows.

• Country concentration: United States is the leading country entry in the split country field.

• Genre mix: International Movies is the most frequent genre entry.

• Rating mix: TV-MA is the most frequent rating label; these are catalogue classifications, not user review scores.

• Trend interpretation: release year and date-added year answer different questions and should not be conflated.

• Data quality: preserve unknown categorical values with an explicit label and validate results whenever the source dataset is refreshed.

Limitations & Submission Notes

Findings describe only the supplied CSV snapshot, not the live Netflix catalogue. Country and genre counts are entry counts because these columns can contain multiple values. The supplied brief says four of six tasks are required; this project covers all six. The PDF brief lists source code, screenshots, a GitHub link if available, and an optional demo video as submission items.

Included files: Python source code, Jupyter notebook, cleaned CSV, summary tables, charts, README, and this report. Run the code locally before submission and review all outputs.